

Thesis

Learning Latent Representations of Brain Dynamics for Modelling Alzheimer's Disease Progression

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Abstract

Acknowledgements

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Introduction

1.1 Context and Motivation

1.2 Objectives and Research Questions

1.3 Document Structure

State of the Art

2.1 Alzheimer's Disease and Biomarkers

Find and reference papers on:

- HC, MCI and AD stages, clinical progression of Alzheimer's disease, motivation for modelling disease-related changes.
- Amyloid- β and tau pathology, relationship with Alzheimer's disease progression, motivation for including biomarker levels as additional biological information.
- Biological definition of Alzheimer's disease using biomarkers, relationship between clinical diagnosis and underlying AD pathology.

2.2 Functional Neuroimaging and Brain Connectivity

rs-fMRI, BOLD, parcellation, FC, dynamic FC/SwFCD, functional networks.

Find papers on:

- BOLD signal and resting-state fMRI, measurement of spontaneous brain activity, basis of the fMRI time-series data used in the thesis.
- Correlation between regional BOLD signals, static functional connectivity, representation of functional relationships between brain regions.
- Time-varying functional connectivity, sliding-window analysis, motivation for evaluating temporal changes in connectivity rather than only static FC.

reference:

- "Local-Global Parcellation of the Human Cerebral Cortex from Intrinsic Functional Connectivity MRI" — Alexander Schaefer et al. (2018). Schaefer cortical parcellation, functionally defined cortical regions, basis of the 400-region representation used for the BOLD signals.
- "The Organization of the Human Cerebral Cortex Estimated by Intrinsic Functional Connectivity" — B. T. Thomas Yeo et al. (2011). Functional organization of the cortex, seven- and seventeen-network parcellations, basis for interpreting latent and decoder differences using the Yeo-7 networks. IF the latent analysis using these networks will be used.

2.3 Representation Learning and Autoencoders

dimensionality reduction, PCA, AE, latent space, VAE.

use papers on autoencoders and ML:

- “Reducing the Dimensionality of Data with Neural Networks” — Geoffrey E. Hinton & Ruslan R. Salakhutdinov (2006). General AE paper, comparison with PCA, deep AEs as a nonlinear dimensionality-reduction approach.
- “Representation Learning: A Review and New Perspectives” — Yoshua Bengio, Aaron Courville & Pascal Vincent (2013). General representation learning review, motivation for learning useful features from high-dimensional data, comparison of representation-learning approaches.
- “Auto-Encoding Variational Bayes” — Diederik P. Kingma & Max Welling (2013/2014). Foundational VAE paper, probabilistic latent representations, reparameterization trick and KL-divergence regularization.
- “Extracting and Composing Robust Features with Denoising Autoencoders” — Pascal Vincent, Hugo Larochelle, Yoshua Bengio & Pierre-Antoine Manzagol (2008). Denoising AE paper, learning robust representations by reconstructing clean inputs from corrupted data, extension of the standard reconstruction objective. MAYBE, as the task used in thesis doesn’t relate to denoising...
- “On Lines and Planes of Closest Fit to Systems of Points in Space” — Karl Pearson (1901). Foundational PCA work, linear dimensionality reduction, useful as the classical statistical comparison to autoencoder-based dimensionality reduction.

2.4 Representation Learning of Functional Brain Dynamics

reference papers:

- “Generative Embeddings of Brain Collective Dynamics Using Variational Autoencoders” — Yonatan Sanz Perl et al. (2020). VAE applied to brain activity, low-dimensional embedding of collective brain dynamics, analysis of meaningful organization within the learned latent space.
- “Representation Learning of Resting State fMRI with Variational Autoencoder” — Jung-Hoon Kim, Yizhen Zhang, Kuan Han, Zheyu Wen, Minkyu Choi & Zhongming Liu (2021). Convolutional VAE applied directly to resting-state fMRI, individual fMRI frames represented in a low-dimensional latent space, analysis of brain activity through learned representations.
- “Low-Dimensional Organization of Global Brain States of Reduced Consciousness” — Yonatan Sanz Perl et al. (2023). VAE applied to parcellated resting-state fMRI, low-dimensional representation of global brain states, comparison of latent organization between different states of consciousness.

- “Modelling Low-Dimensional Interacting Brain Networks Reveals Organising Principle in Human Cognition” — Yonatan Sanz Perl et al. (2025). Low-dimensional modelling of fMRI brain activity, interpretation of latent variables through functional brain networks, analysis of interactions and coupling between networks in the latent representation.

reference papers where other types of AEs are applied.

2.5 Autoencoder-Based Representation Learning in Alzheimer’s Disease

reference papers:

- “Deep Learning-Based Feature Representation for AD/MCI Classification” — Heung-Il Suk & Dinggang Shen (2013). Deep learning for AD/MCI representation learning, extraction of disease-relevant features from neuroimaging/biomarker data, use of learned representations for diagnostic classification.
- “Reducing Variations in Multi-Center Alzheimer’s Disease Classification with Convolutional Adversarial Autoencoder” — Bernard M. Cobbinah et al. (2022). Convolutional adversarial AE for Alzheimer’s classification, learning disease-relevant representations while reducing differences caused by multi-center data, downstream diagnostic classification.
- “Automated Prediction System for Alzheimer Detection Based on Deep Residual Autoencoder and Support Vector Machine” — M. Menagadevi, S. Mangai, Nirmala Madian & D. Thiyagarajan (2023). Deep residual AE for Alzheimer’s feature extraction, dimensionality reduction followed by SVM classification, use of AE representations for automated AD detection.
- “Latent Diffusion Autoencoders: Toward Efficient and Meaningful Unsupervised Representation Learning in Medical Imaging – A Case Study on Alzheimer’s Disease” — Gabriele Lozupone, Alessandro Bria, Francesco Fontanella, Frederick J. A. Meijer, Claudio De Stefano & Henkjan Huisman (2026). Latent autoencoder representation learning for medical imaging, Alzheimer’s disease case study, focus on learning compact and meaningful unsupervised representations.
- “Using Deep Autoencoders to Identify Abnormal Brain Structural Patterns in Neuropsychiatric Disorders: A Large-Scale Multi-Sample Study” — Walter H. L. Pinaya, Andrea Mechelli & João R. Sato (2018). Deep AE applied to structural brain imaging, learning normative representations of healthy brain structure, use of reconstruction differences to identify abnormal patterns in clinical populations. MAYBE as it is not just AD but other disorders too.

2.6 Longitudinal and Disease-Progression Modelling

reference:

- “Disentangled Sequential Graph Autoencoder for Preclinical Alzheimer’s Disease Characterizations from ADNI Study” — Fan Yang, Rui Meng, Hyunwoo Cho, Guorong Wu & Won Hwa Kim (2021). Sequential graph AE for Alzheimer’s disease characterization, modelling relationships between brain regions and longitudinal clinical visits, learning representations of disease-related changes over time.
- “Model-Based Whole-Brain Perturbational Landscape of Neurodegenerative Diseases” — Yonatan Sanz Perl et al. (2023). Whole-brain computational modelling of neurodegenerative diseases, analysis of disease-specific alterations in brain dynamics, investigation of how perturbations can move diseased brain dynamics toward healthier states. MAYBE as it is not only for AD but other diseases as well.

2.7 Brain Connectivity Representation and Translation

reference:

- “Krakencoder: A Unified Brain Connectome Translation and Fusion Tool” — Keith W. Jamison, Zijin Gu, Qinxin Wang, Ceren Tozlu, Mert R. Sabuncu & Amy Kuceyeski (2025). Encoder-decoder framework for brain connectomes, shared latent representation of different connectivity modalities, reconstruction and translation between brain connectivity representations.
- “Modelling Low-Dimensional Interacting Brain Networks Reveals Organising Principle in Human Cognition” — Yonatan Sanz Perl et al. (2025). Low-dimensional representation of functional brain networks, latent-space interpretation of network interactions, relevant to analysing how latent dimensions relate to functional brain organization.

Methodology

3.1 Data and Preprocessing

3.1.1 ADNI Dataset

Copy pasted from the longitudinal dataset description, since they stated it follows the same processing as ADNI-B. Needs to be heavily adjusted

We have preprocessed the functional MRI data following the standard volume-based pipeline implemented within the Matlab-based CONN toolbox (<http://www.nitrc.org/projects/conn>), version 22v2407. The following steps are included in the standard CONN pipeline: functional realignment and motion correction, slice-timing correction to account for differences in time of acquisition between slices, identification of outlier scans for subsequent regression by means of the quality assurance/artifact rejection software Artifact Detection Toolbox (ART, http://www.nitrc.org/projects/artifact_detect), coregistration with each participant's high-resolution T1-weighted image, spatial normalisation to MNI-152 standard space with 2mm isotropic resampling resolution and smoothing with a Gaussian kernel of 8mm FWHM. Denoising also follows CONN's denoising approach: to reduce noise due to cardiac and motion artifacts we applied the anatomical CompCor method (4) (also implemented within the CONN toolbox), by regressing out of the functional data the first five principal components attributable to each individual's white matter signal, the first five components attributable to individual cerebrospinal fluid (CSF) signal, six subject-specific realignment parameters (three translations and three rotations) as well as their first-order temporal derivatives, the artifacts identified by ATRT, and main effect of scanning condition (None in this case). In the same denoising step, each individual's denoised BOLD signal timeseries is band-pass filtered in the 0.008-0.08 Hz range, to eliminate both low-frequency drift effects and high-frequency noise.

Talk about the actual format of the parcelated data, distribution of classes, other data used like the Amyloid Beta and Tau levels, how they were normalised and what was done for subjects that didn't have this data.

3.1.2 Longitudinal ADNI Dataset

Although the longitudinal dataset was intended to undergo the same preprocessing pipeline as the ADNI dataset, practical differences remained between the two resulting datasets. Comparing the BOLD time series for sessions shared by the same subjects shows that the signals differ slightly, as illustrated in Figure 3.1. The mean Pearson correlation for

sliding-window functional connectivity dynamics (SwFCD) is 0.533, while the corresponding functional connectivity (FC) Pearson correlation is 0.7019. Consequently, model-evaluation results obtained with the longitudinal dataset should be expected to have similar values.

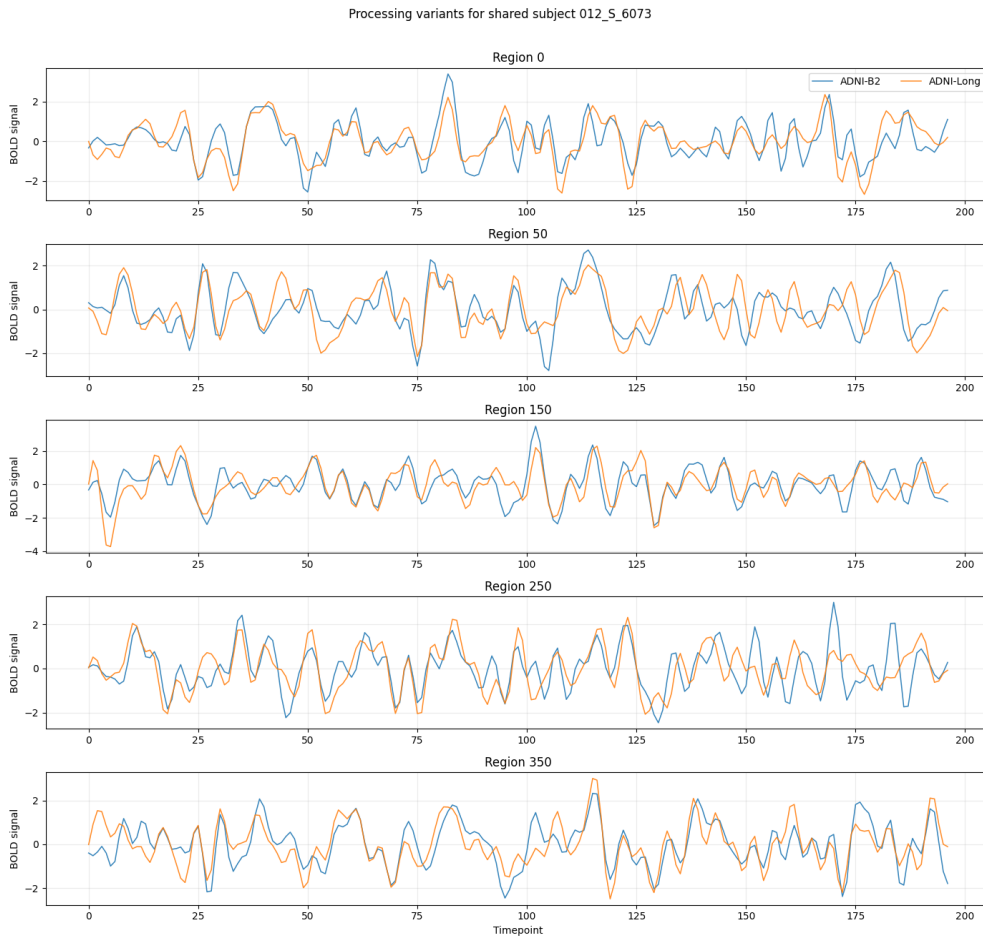


Figure 3.1: Comparison of BOLD time series of selected regions 0, 50, 150, 250, 350 from a common session of subject 012_S_6073 in the ADNI and longitudinal ADNI datasets. Despite theoretically identical preprocessing, small practical differences are present.

The SwFCD and FC matrices offer a complementary comparison of the two datasets displayed in Figure 3.2...

finish

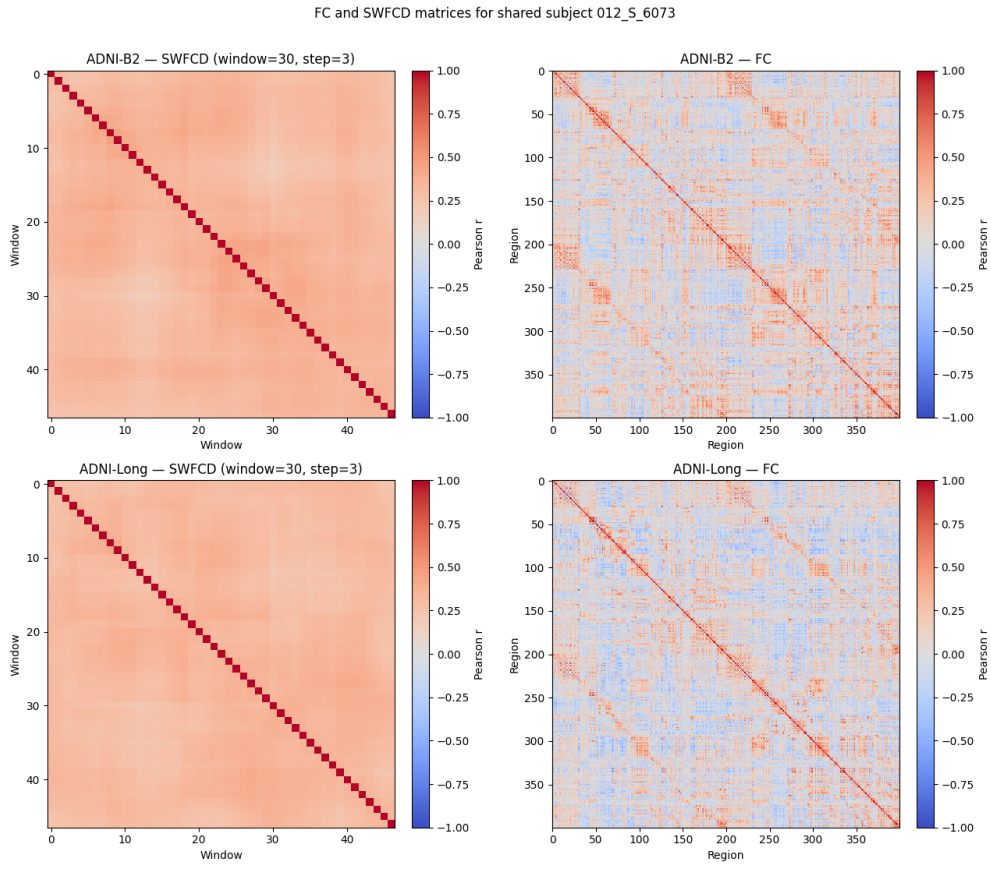


Figure 3.2: Comparison of SwFCD and FC matrices for subject 012_S_6073 between the ADNI and longitudinal ADNI datasets.

Add more on how the dataset was split into pairs, i.e. prepared for evaluation.

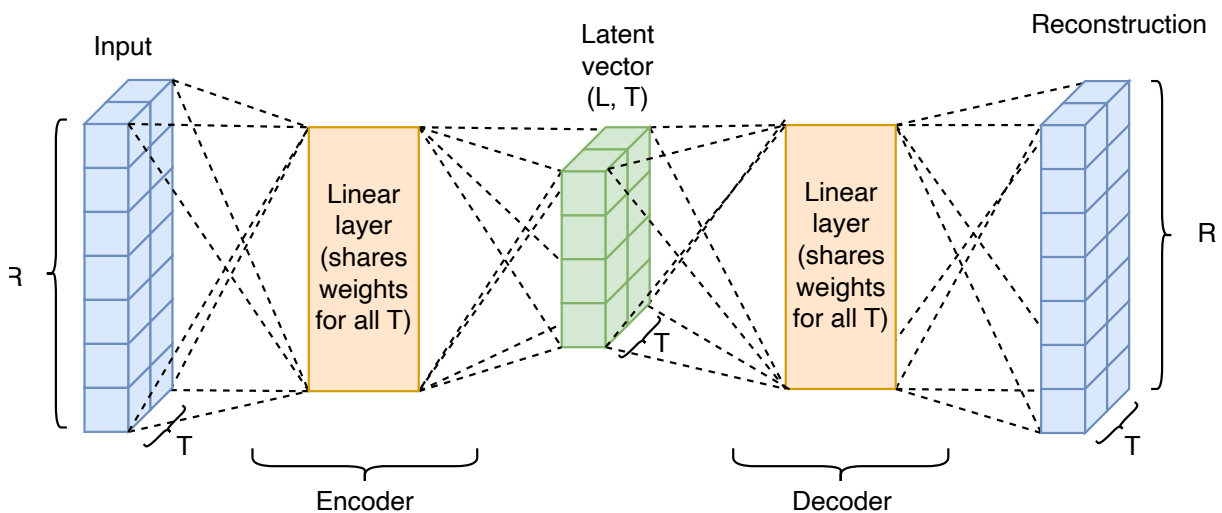
3.1.3 Additional data processing

Talk about how both of the datasets were normalised using z-score across subjects and regions, maybe move this section under the "ADNI Dataset" as it talks mostly about processing...

3.2 Autoencoder Framework

General encoder-latent-decoder formulation. Input and output representation: regions by timepoints. Latent-space dimensionality. How the spatial/regional dimension is compressed while temporal structure is retained.

3.2.1 Linear Autoencoder



Consists of one linear layer in both the encoder and decoder, reducing the input regions to latent dimension and from latent back to the same initial number of regions.

This architecture has only one variable parameter which is the latent dimension.

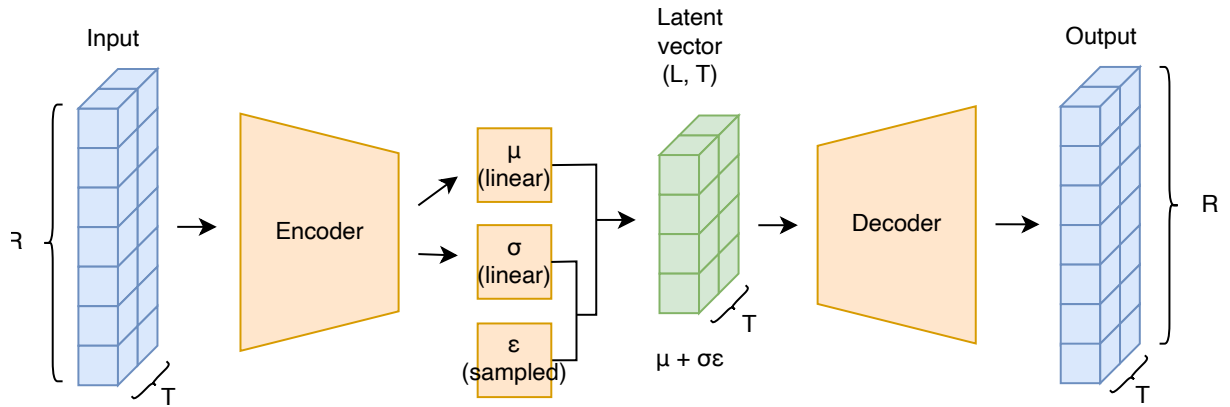
Reference architecture graphic

3.2.2 Deep Autoencoder

3.2.3 Convolutional Autoencoder

3.2.4 Variational Autoencoder Extension

Talk about the KL-divergence regularisation term added to each of the architectures above, where the deterministic latent representation becomes a probabilistic latent distribution.



3.3 Training Objectives

Talk about:

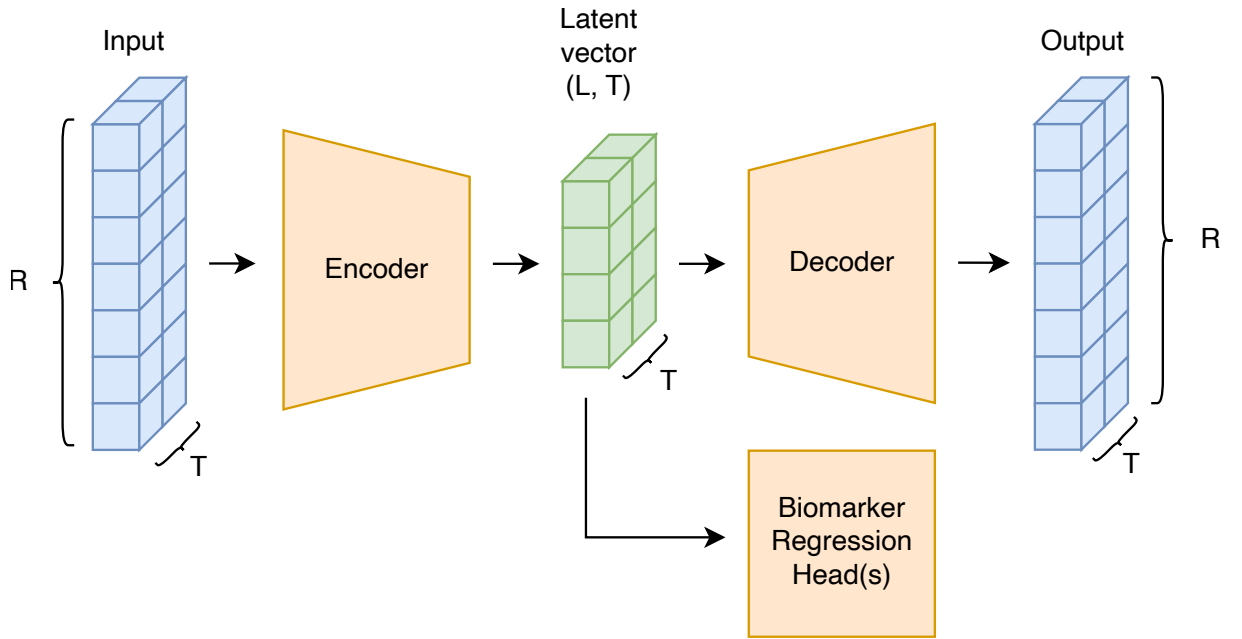
- Reconstruction MSE.
- Functional connectivity loss.
- How FC matrices are calculated for the loss.
- temporal derivative loss.
- Dynamic/SwFCD-related variability loss or surrogate.
- Motivation for each loss: what property of the reconstructed BOLD signal it attempts to preserve.
- Combined reconstruction objective.
- Weighting coefficients for individual losses
- KL term for variational models.

3.4 Auxiliary Supervision

Mainly about adding classification and regression heads to predict diagnostic, amyloid, and tau levels to see if it improves reconstruction or the latent space. In detail:

- Motivation for encouraging disease-relevant information in the latent space.
- Diagnostic classification head.
- HC/MCI/AD classification target.
- Classification-head architecture.
- Classification loss.
- Handling of class imbalance, if applicable.

- Amyloid- β regression head.
- Tau regression head.
- Whether biomarkers are predicted region-wise or globally.
- Biomarker target preprocessing/normalisation, dealing with missing data.
- Regression loss.
- Combined multi-task objective.
- Weighting between reconstruction, classification and regression losses.
- Which parts of the network receive gradients from auxiliary tasks.



3.5 Class-Specific Decoder Framework

talk about

- Motivation for separating shared representation learning from disease-specific decoding.
- General encoder E , General decoder D_G .
- HC-specific decoder D_{HC} , MCI-specific decoder D_{MCI} , AD-specific decoder D_{AD}
- Use of the same latent representation for every decoder.
- Which components are frozen during class-specific decoder training.
- Initialization of class-specific decoders (e.g. from the general decoder, if applicable).
- Training data used for each decoder.

- Loss used to train the class-specific decoders.
- Interpretation of decoder differences as disease/class-specific mappings rather than differences caused by separate encoders.
- Application of target-class decoders to longitudinal data.
- Earlier scan as input and later scan as the longitudinal target.
- HC→MCI, MCI→AD, etc. transitions.

3.6 Evaluation Metrics

subsections:

- MSE, RMSE
- FC Pearson
- SwFCD Pearson

3.7 Interpretability Methods

try to do

- latent space analysis
- Latent-to-brain correspondence, how each latent dimension is associated with each Schaefer parcel.
- Network-level interpretation. Mapping Schaefer parcels to Yeo-7 networks, then aggregating the associations to those networks.
- decoder comparison

Experimentation and Evaluation

4.1 Experimental Setup

Number of epochs for each model determined using the k-fold cross validation method with 5 folds.

- 4.2 Autoencoder Architecture Selection**
 - 4.2.1 Linear AE Parameter Search**
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- 4.3 Baseline Model Characterisation and Analysis**
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 - 4.4.3 Auxiliary Diagnostic Classification**
 - 4.4.4 Amyloid and Tau Auxiliary Regression**
 - 4.4.5 Training Hyperparameter Optimisation**
 - 4.4.6 Final Model Selection**
- 4.5 Final Model and Latent Representation Analysis**
- 4.6 Class-Specific Decoder Experiments**
 - 4.6.1 Decoder Training and Performance**
 - 4.6.2 General versus Class-Specific Decoders**
 - 4.6.3 Shared Latent-to-Brain Organisation**
- 4.7 Longitudinal Evaluation**

Discussion

Conclusion

Appendices

A.1 Supplementary Material